

Vulnerability Assessment Report

May 12, 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The assessment will cover a period of three months, from February 2026 to April 2026.

Purpose

The employees of this company work remotely across different geographical locations, meaning having access to this remote server is essential for business operations. The server contains customer information that needs to be readily available for an employee to query or request. Due to the fact the database is publicly accessible, it's imperative to secure the data to ensure its integrity will not be compromised. If the server were to be disabled in any way, this would result in the business suffering major financial repercussions and disruption in business operations.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Obtain sensitive information via exfiltration	1	3	3
Networking	DoS/DDoS disrupts server connectivity	2	2	4
Power Outage	A major power outage shuts down the servers environment	1	2	2

Approach

The exfiltration of sensitive information can have serious consequences for a business's reputation among customers, potentially leading to significant financial losses. Social engineering and phishing attacks are common threats today and can be mitigated through proper awareness training, implementation of access control groups, and enforcement of the principle of least privilege.

A DoS attack targeting database servers, or a power outage, could result in services becoming temporarily unavailable to customers. This can lead to financial impact if appropriate disaster recovery plans and risk mitigation measures are not in place.

Remediation Strategy

To reduce the risk of data exfiltration, phishing, and unauthorized access, implement strong authentication, authorization, and auditing controls on the database server. This includes enforcing multi-factor authentication, strong password policies, and role-based access control to ensure users only have the minimum required privileges. User awareness training should be conducted regularly to help reduce successful social engineering and phishing attempts.

To improve resilience against service disruption from DoS attacks or power outages, establish disaster recovery and continuity plans, including regular backups and failover systems where possible. Network-level protections such as IP allow-listing (e.g., restricting database access to corporate offices or trusted networks) should also be implemented to reduce exposure to external threats.

Document Control

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